Tim S. Haubold

Curriculum Vitae

timshaubold.github.io

Education

2023 **PhD**, *Institute of Applied Mathematics, University of Hannover*, *Summa Cum Laude*, Thesis title: *Applications of Special Functions in High Order Finite Element Methods* (Advisor: Prof. Dr. Sven Beuchler)
Keywords: Numerical analysis, orthogonal polynomials, symbolic computation, high order finite elements

2017 **B.Sc and M.Sc**, *University of Bremen*, Master thesis title: *Hydrostatic reconstructed FORCE scheme and application to the non-unique Riemann solutions for the shallow water equations* (Advisor: Prof. Dr. Alfred Schmidt)
Keywords: Finite Volumes, shallow water equations

Employment

2026 **Postdoctoral researcher**, *École nationale des ponts et chaussées / Inria Paris - Team SERENA with Alexandre Ern*
Keywords: Unfitted Finite Element, Time-dependent problem, Wave equation

2023–2026 **Postdoctoral researcher**, *Georg-August-Universität Göttingen with Christoph Lehrenfeld as main collaborator*
Keywords: Unfitted Finite Elements, Maxwell's equation, Numerical Analysis

2017–2023 **Predoctoral researcher**, *Gottfried-Wilhelm-Leibniz-Universität Hannover under supervision of Sven Beuchler*
Keywords: Symbolic Computation, High order finite element, Basis construction

2014–2017 **Student Assistant**, *Universität Bremen*, SFB 747: Mikrokaltumformen

Publications (Main author)

2024 **High order biorthogonal functions in** $H(\mathrm{Curl})$, *Haubold, Beuchler, Schöberl*, SIAM Journal on Numerical Analysis, Volume 62, Nr. 5, Pages 2331–2348
https://doi.org/10.1137/23M1606794

2024 **Recursion Formulas for Integrated Products of Jacobi Polynomials**, *Beuchler, Haubold, Pillwein (alphabetical)*, Constructive Approximation, Volume 59, Pages 583–618
https://doi.org/10.1007/s00365-023-09655-z

2022 **Recurrences for Quadrilateral High-Order Finite Elements**, *Beuchler, Haubold, Pillwein (alphabetical)*, Mathematics in Computer Science, Volume 16, article number 32
https://doi.org/10.1007/s11786-022-00547-2

Proceedings

2023 **Applications of Special Functions in High Order Finite Element Methods**, *Haubold, Beuchler, Pillwein, Schöberl*, Mini-Workshop: Multivariate Orthogonal Polynomials: New synergies with Numerical Analysis, Workshop Report 2023,43
https://doi.org/10.14760/OWR-2023-43

2021 **Recursion relations for hp-FEM Element Matrices on quadrilaterals**, *Haubold, Pillwein, Beuchler*, PAMM, Volume 21, Issue 1, e202100200
https://doi.org/10.1002/pamm.202100200

2019 **Symbolic Evaluation of hp-FEM Element Matrices**, *Haubold, Pillwein, Beuchler*, PAMM, Volume 19, Issue 1, e201900446
https://doi.org/10.1002/pamm.201900446

Preprints

2026 **High order biorthogonal functions for the discrete de Rham complex on simplices**, *Haubold, Beuchler, Schöberl*, arXiv:2609.28345
https://arxiv.org/abs/2609.28345

In preparation (Main author)

2026 **Parameter-robust unfitted finite element methods for a Maxwell interface problem**, *Haubold, Lehrenfeld*

2026 **Optimal decay rate of high order local operator**, *Haubold, Maier*

2026 **Unfitted discontinuous Galerkin methods for the linear wave equation**, *Ern, Haubold (alphabetical)*

Teaching

2026 **Exercises (English)**, *Éléments finis*

2025-2026 **Lecture and Exercise**, *Mathematisch orientiertes Programmieren*, Joint with Dr. Igor Voulis

2025 **Exercises**, *Functional Analysis*

2024-2025 **Lecture and Exercises**, *Numerics of partial differential equations III*, Joint with Prof. Dr. Christoph Lehrenfeld

2024 **Lecture**, *Mathematisch orientiertes Programmieren*, Joint with Dr. Igor Voulis

2023-2024 **Exercises**, *Numerics of partial differential equations I and II*

2017-2023 **Exercises**, *Numerics I, Numerics II, Numerics of partial differential equations (german and english), Numerical methods for continuum mechanics, Numerics for teachers, mathematics for special-needs teachers, mathematics for civil engineers, . . .*, in average 4 hours at the blackboard

2022 **Lecture and Exercise**, *Interaktive algorithmische Einführung in die Finiten Elemente*, Examiner: Prof. Dr. Sven Beuchler

2020 **External Lecture at Leibniz Fachhochschule Hannover**, *Statistics I*

Research visits

19.-23.11.18 **JKU Linz**, *Linz, Austria*, Working with Veronika Pillwein
14.-15.09.23 **TU Wien**, *Vienna, Austria*, Working with Joachim Schöberl
15.10.25 **TU Hamburg**, *Hamburg, Germany*, Invited seminar talk
09.-12.12.25 **KIT**, *Karlsruhe, Germany*, Working with Roland Maier
01.10.26 **LmB**, *Besançon, France*, Invited seminar talk

Talks

2026 **GAMM Annual Meeting**, *Stuttgart, Germany*, Co-Chair of Section 18
2025 **International Conference on Domain Decomposition Methods**, *Milan, Italy*
2025 **GAMM Annual Meeting**, *Poznan, Poland*
2025 **SIAM CSE**, *Fort Worth, USA*
2024 **Chemnitz Finite Element Symposium**, *Chemnitz, Germany*
2024 **WCCM-PANACM**, *Vancouver, Canada*
2024 **ECCOMAS Congress**, *Lisboa, Portugal*
2024 **GAMM Annual Meeting**, *Magdeburg, Germany*
2023 **Oberwolfach Mini-Workshop: Multivariate Orthogonal Polynomials: New synergies with Numerical Analysis**, *Oberwolfach, Germany*
2023 **Chemnitz Finite Element Symposium (on tour)**, *Leibnitz, Austria*
2022 **WAND**, *Salzburg, Austria*
2022 **GAMM Annual Meeting**, *Aachen, Germany*
2021 **ICOSAHOM**, *Vienna, Austria (online)*
2021 **GAMM Annual Meeting**, *Kassel, Germany (online)*
2019 **Chemnitz Finite Element Symposium (on tour)**, *Essen, Germany*
2019 **GAMM Annual Meeting**, *Vienna, Austria*
2019 **GAMM-Workshop 2019 - "Young Academics Workshop"**, *Sion, Switzerland*
2018 **Chemnitz Finite Element Symposium**, *Chemnitz, Germany*

Skills

Finite element software: NGSolve, ngsxfem
Programming: Python, MATLAB, C++ (basic)
Computer algebra: Mathematica

Languages

German native speaker
English fluent

Additional Information

2026 **Co-Chair of Section 18 at the GAMM Annual Meeting**

2024 **FE Stokes Re-Pair (a nerdy card game)**, *Haubold, Lehrenfeld, van Beeck, Zerbinati, Zienecker*
https://fe-nerd-games.github.io/FEStokesRePair/

2022-2023 **Employee representative and board member of the Institute of Applied Mathematics, University of Hannover**

2020-2022 **Employee vice representative of the Institute of Applied Mathematics, University of Hannover**

2023 **Local organizer of the first internal poster day at the IfAM, University of Hannover**

2022 **Part of the local organizing team of the annual Northern German Colloquium (NoKo) 2022**

References

Sven Beuchler, *PhD - Advisor*

Alexandre Ern, *Postdoctoral mentor*

Christoph Lehrenfeld, *Postdoctoral mentor*

Joachim Schöberl, *Collaborator*

Collaborators

Sven Beuchler, *Leibniz University Hannover, DE*

Veronika Pillwein, *Johannes Kepler University Linz, AT*

Joachim Schöberl, *TU Wien, AT*

Christoph Lehrenfeld, *University of Göttingen, DE*

Roland Maier, *Karlsruher Institut für Technologie, DE*

Alexander Heinlein, *TU Delft, NL*

Alexandre Ern, *ENPC Paris / Inria Paris, FR*